

Research Associate at NASA Goddard/University of Maryland | Polar Engineer/Glaciologist
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APPOINTMENTS

Research Associate, Earth System Science Interdisciplinary Center,
NASA Goddard Space Flight Center and University of Maryland, USA
Research Associate, Dartmouth College, USA

Sep 2026 - Present
Oct 2025 - Aug 2026

EDUCATION

Doctor of Philosophy

Aug 2021 - Sep 2025

Dartmouth College, Hanover, NH

Thesis: *Old, Flat, and Slow: Interior Greenland Snow and Ice Dynamics Revealed with GNSS*

Hardware and software designs of low-power, scientific-grade GNSS: github.com/glaciology/OGRE

Advisor: Dr. Robert L. Hawley

- Lead for various GNSS field projects, including validating NASA laser altimetry measurements, studying crevasse formation, improving firn densification models, and improving space-based ice velocity estimates.
- Designed OGRE (Open GNSS Research Equipment), a field-deployable GNSS instrument for polar research, reducing costs by 44-90% compared to commercial alternatives
- Optimized OGRE for extreme environments, achieving best-in-class low power consumption, minimizing battery requirements for long-term, over-winter autonomous deployments in remote field settings.
- Developed an autonomous radar technique for monitoring snow surface roughness, enhancing the temporal and spatial resolution of data in understudied polar regions.
- Deployed and supported over 100 OGRE units across Greenland, the USA, and Antarctica as part of 7+ funded research projects, supporting studies in geophysics, hydrology, crevassing, and ice sheet dynamics.
- Accumulated over 52 weeks of field experience in Arctic and Antarctic environments, demonstrating expertise in remote field operations, instrument deployment, and extreme environment logistics.

Bachelor of Arts in Physics - Applied Track

June 2020

Pomona College, Claremont, CA

Thesis: *Computer Vision Software Development for an Explosive Seed Pod Trap Camera*

Engineering Capstone: *Position Determination from Star Tracker Earth Observations*

University of Cambridge, Cambridge U.K. (2019): *Characterizing Chemical Abundances of Select A-type Stars*

PUBLISHED PAPERS

- [5] **Pickell, D. J.**, Hawley, R. L., Good, J. C. ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. *The Cryosphere*. 2026.
- [4] **Pickell, D. J.**, Hawley, R. L., Lewinter, A. Spatiotemporal patterns of accumulation and surface roughness in interior Greenland with a GNSS-IR network. *The Cryosphere*. 2025.
- [3] Wilner, J. A., Nordin, B. J., Getraer, A., Gregoire, R. M., Krishna, M., Li, J., **Pickell, D. J.**, Rogers, E. R., McDannell, K. T., Palucis, M. C., Keller, C. B. Limits to timescale dependence in erosion rates: Quantifying glacial and fluvial erosion across timescales. *Science Advances*. 2024.
- [2] Khan, S. A., Morlighem, M., Ehrenfeucht, S., Seroussi, H., Choi, Y., Rignot, E., Humbert, A., **Pickell, D. J.**, Hassan, J. Inland summer speedup at Zachariæ Isstrøm, northeast Greenland, driven by subglacial hydrology. *Geophysical Research Letters*. 2024.
- [1] **Pickell, D. J.**, Hawley, R. L. Performance characterization of a new, low-cost multi-GNSS instrument for the cryosphere. *Journal of Glaciology*. 2024.

PAPERS IN REVIEW & MANUSCRIPTS

- *In Review March 2026.* **Pickell, D. J.**, Hawley, R. L., Stevens, C. M. A single instrument approach to ice flow, surface mass balance, and firn densification validation in central Greenland.
- *In Review April 2026.* Sledd, A., Gallagher, M. R., Shupe, M. D., Cox, C. J., Hawley, R. L., Town, M. S., Guy, H., Marshall, H.-P., Neely, R. R., III, Pettersen, C., Walden, V. P., Hebson, C., Martin, A., Olson, E., **Pickell, D. J.** Summer subsurface temperature variability in the percolation zone of southwest Greenland: high resolution observations of the top meter of firn. EGU sphere [preprint]. <https://doi.org/10.5194/egusphere-2026-1842>. 2026.
- *In preparation.* Gallagher, M. R., Shupe, M. D., Walden, V. P., Sledd, A., Cox, C. J., Guy, H., Pettersen, C., Martin, A. S., Neely, R. R., III, Hawley, R. L., Town, M. S., Olson, E., Marshall, H.-P., **Pickell, D. J.**, Vandecrux, B., Mottram, R., Mattingly, K. S., Clemens-Sewall, D., Hebson, C. How Greenland melts under cloudy and clear skies. Planned submission: April 2026.
- *In preparation.* **Pickell, D. J.**, Scott, J. GNSS interferometric reflectometry reveals detailed lake surface phenology. Planned submission: mid 2026.
- *In preparation.* **Pickell, D. J.** Widespread changes in Greenland surface roughness influences how it melts. Planned submission: mid 2026.

CONFERENCE ABSTRACTS, PRESENTATIONS, & INVITED TALKS

- [12] **Pickell, D. J.** ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. NASA ICESat-2 Science Project Office talk. 2026.
- [11] Togaibekov, A., Khan, S. A., Løkkegaard, A., Colgan, W., **Pickell, D. J.** Temporal evolution and kinematics of a mega-crevasse at Jakobshavn Isbræ revealed by dense GNSS observations. EGU General Assembly 2026.
- [10] **Pickell, D. J.**, Hawley, R. L., Good, J. C. ICESat-2 surface elevation assessment with kinematic GPS and static GNSS near the ice divide in Greenland. The Cryosphere. 2026.
- [9] Wilner, J. A., Nordin, B. J., Getraer, A., Gregoire, R. M., Krishna, M., Li, J., **Pickell, D. J.**, Rogers, E. R., McDannell, K. T., Palucis, M. C., Keller, C. B. Global quantification of glacial versus fluvial erosion rates: limits to timescale dependence. *EGU General Assembly 2025*.
- [8] Løkkegaard, A., Colgan, W., Khan, S. A., Fahrner, D. R., Polzin, M., Hughes, J., Lippert, E. Y., **Pickell, D. J.** Brittle landscapes: A case study of crevasse development in firn. *EGU General Assembly 2025*.
- [7] **Pickell, D. J.**, Hawley, R. L. Spatio-temporal patterns of surface accumulation and surface roughness in the interior region of Greenland with a novel GNSS network. *AGU Fall Meeting 2024*.
- [6] Gallagher, M. R., Walden, V. P., Shupe, M. D., Guy, H., Martin, A. S., Sledd, A., Pettersen, C., Neely, R. R., III, Marshall, H.-P., Hawley, R. L., Olson, E., Mattingly, K. S., Hebson, C., Neff, W. D., **Pickell, D. J.**, Town, M. S. How Greenland melts: high fidelity observations of the coupled climate system in Southwestern Greenland from summer 2024. *AGU Fall Meeting 2024*.
- [5] **Pickell, D. J.**, Hawley, R. L. Surface mass balance and elevation change in Greenland with a network of low-cost GNSS stations. *AGU Fall Meeting 2023*.
- [4] Good, J. C., Hawley, R. L., **Pickell, D. J.** Advancing snow surface roughness observations of the Greenland Ice Sheet using down-looking laser distance meters. *AGU Fall Meeting 2023*.
- [3] **Pickell, D. J.**, Hawley, R. L. Altimetry validation and snow surface monitoring with an on-ice array of open source multi-GNSS instruments. *AGU Fall Meeting 2022*.
- [2] **Pickell, D. J.**, Hawley, R. L. An on-ice GNSS Research Experimental Network for Greenland (OGRENet). *Northeastern Glaciology Meeting 2021*.
- [1] **Pickell, D. J.**, Hawley, R. L. An on-ice GNSS Research Experimental Network for Greenland (OGRENet). *AGU Fall Meeting 2021*.

PUBLISHED DATASETS

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- [4] **Pickell, D. J.**, Hawley, R. L., Brunt, K. M., Dorsi, S. W., McConnell, J. R., Neumann, T. A. & Pettit, J. ICESat/ICESat-2 Traverse: Monthly GPS Surface Elevation Data at Summit Station, Greenland. (IS2TGPSSS, Version 1). [Data Set]. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. 2026.
- [3] Hawley, R. L., **Pickell, D. J.**, McConnell, J. R., Neumann, T. A., Felikson, D. & Dorsi, S. W. Weekly snow stake array measurements, Summit Station, Greenland (IS2SSAGCV, Version 1). Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. 2026.
- [2] **Pickell, D. J.**, Hawley, Robert L., Brunt, Kelly M., Burkhart, John F., Dorsi, Samuel W., McConnell, Joseph R., Neumann, Thomas A., Pettit, Joe. ICESat/ICESat-2 traverse: monthly GPS surface elevation data at Summit Station, Greenland. Dartmouth Dataverse 2025.
- [1] **Pickell, D. J.**, R. L. Hawley. An East-West Network of Twelve Global Navigation Satellite System (GNSS) Stations in the Summit Region of Greenland: RINEX Data, 2022-2025. Arctic Data Center. 2024.

SIGNIFICANT FIELD EXPERIENCE

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| Summit, Greenland, <i>Expedition Lead</i> , snow and ice dynamics, 2 weeks | 2026 |
| Juneau Icefield, <i>Teaching Faculty</i> , GNSS and glaciology, 2 weeks | 2026 |
| Drang Drung Glacier, India, <i>Expedition Member</i> , ground radar (radio echo), 1 week | 2026 |
| McMurdo/Ross Ice Shelf Shear Zone, <i>Expedition Member</i> , ground penetrating radar, 4 weeks | 2025 |
| Qaanaaq, Greenland, <i>Expedition Member</i> , GNSS instrument deployment, 3 weeks | 2025 |
| Summit, Greenland, <i>Expedition Lead</i> , snow and ice dynamics, 2 weeks | 2025 |
| Athabasca Glacier, Canada, <i>Team Instructor/Leader</i> , glaciology field education program, 1 week | 2022, 2024, 2025 |
| Leverett Glacier, Antarctica, <i>Radar Engineer + Mountaineer Safety</i> , Crevasse mapping, 4 months | 2024 |
| Mega-crevasse Zone, Ilulissat, Greenland, <i>Expedition Member</i> , GNSS instrument deployment, 2 weeks | 2024 |
| Point 660, Greenland, <i>Educational Lead</i> , ice dynamics and mapping, 3 weeks | 2024 |
| Summit, Greenland, <i>Expedition Lead</i> , snow and ice dynamics, 2 weeks | 2024 |
| Summit, Greenland, <i>Science Technician</i> , meteorology, geodesy, glaciology, 9 months | 2021, 2023 |
| North East Greenland Ice Stream, Greenland, <i>Expedition Member</i> , ice dynamics, 1 week | 2023 |
| Summit, Greenland, <i>Expedition Lead</i> , snow and ice dynamics, 2 weeks | 2023 |
| Summit, Greenland, <i>Expedition Member</i> , surface processes/instrument deployment, 2 weeks | 2022 |
| Wilderness First Responder training | 2016, 2022, 2025 |

TEACHING, MENTORSHIP & OUTREACH

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| Juneau Icefield Research Program, Volunteer Teaching Faculty, Juneau Icefield, AK | Summer 2026 |
| Leh Indian Glaciology Summer School - Guest Lecturer | Summer 2026 |
| EARS46 - Earth Science Field Study Program, Athabasca Glacier | Fall 2024 |
| <ul style="list-style-type: none"> Led GNSS, glacier flow, and glacier travel/safety exercises in the Columbia Ice Field | |
| Joint Science & Education Project Graduate Teaching Fellow, Greenland | Summer 2024 |
| <ul style="list-style-type: none"> 3 week Arctic field school in Greenland for Danish, American and Greenlandic high school students Taught basic glacier dynamics and mass balance module and supervised student project on radar and GNSS ice measurements | |
| 'Geolunch' Weekly Science Seminar Coordinator | Fall 2022 |
| EARS46 - Earth Science Field Study Program | Spring 2022 |
| EARS01 - How the Earth Works | Fall 2021 |

MENTORSHIP & ADVISING

- Evan Barrett, *Thesis Committee Member*
- Maia Crichlow, *Advisor, Senior Capstone Project*
- Jacob Gordon, *Advisor, Senior Capstone Project*
- Laura Wilson, *Mentor/Field Lead, Summit Greenland*
- Jadin Scott, *Advisor, Research Assistant*

FUNDING, GRANTS & AWARDS

Pending Support

- Submitted March 2026 (Co-PI). *Study of Height, Accumulation, Roughness, and Densification at Summit (SHARDS)*. NASA ICESat-2 Project Office. \$150,680

Funded Support

- *Geophysical Network Of MEMS for Englacial Sensing (GNOMES)*. Arete Glacier Initiative Foundation. (Lead PI) \$191,622.88
- Evans Family Cluster Funding (Dartmouth College): *OGREs for Open GNSS* (2025) \$23,000.00
- Navigating the New Arctic (NNA) Arctic Observing Technologies Research Award (2025) \$1690.00
- Joint Science and Education Program Fellowship (2024) \$4000.00
- The Arthur L. Irving Institute for Energy & Society Independent Research Grant (2024) \$3000.00
- Dartmouth College Guarini Graduate Fellowship (2021 - 2025) \$88,164.00/yr

Other Funding/Awards

- Earle S. Lenker 1956 Graduate Award for outstanding academic achievement by a graduate student 2026

CONSULTING

- Summit Station Science Coordination Office (SCO) 2024 - present
 - *Member of 5 person SCO, which advises the National Science Foundation and Arctic logistics contractors on various ongoing projects in Greenland in which science operations and research is a stakeholder.*
- Faculty Search Committee Fall 2024
- Reviewer, *Journal of Glaciology, IEEE Transactions on Geoscience and Remote Sensing*

WORK EXPERIENCE

Greenland Science Technician, Polar Field Services Jan 2023 - Jun 2023; Apr 2021 - Jun 2021

- Ensured continuous operation of 15+ scientific instruments in an extreme polar environment, enabling uninterrupted atmospheric and geophysical monitoring for global research initiatives.
- Led field operations for 7 NASA ground data collection traverses to validate the NASA Ice, Cloud, and land Elevation Satellite (ICESat-2), directly contributing to improved satellite-based ice sheet mass balance models.
- Upgraded UNAVCO GPS infrastructure, improving precision of geodetic measurements used for tectonic studies and glacier movement tracking.
- Developed and executed action plans to support field research teams and maximize ground-time efficiency, including radio neutrino installation, ice core drilling, and space probe testing.

Loon (Google X) Flight Assembly Specialist, Polar Field Services August 2020 - February 2021

- Carried out assembly, integration, software/electrical testing and quality control of Loon flight systems at field operations facilities, leading to 30+ balloon launches in a three month period.
- Wrote test plans, performed outgoing preflight certification checks and executed launches with launch team.

- Triaged flight system test failures with engineering team while documenting bugs and process improvements.

Research Engineer: Millennium Space Systems

September 2019 - May 2020

- Led a technical trade study on optical-based positioning methods as part of a Harvey Mudd College Clinic team, identifying system improvements to enhance satellite navigation and autonomy.
- Developed and implemented image-processing algorithms and software that expanded the satellite star tracker's capabilities, enabling multi-source navigation using Earth features, increasing operational time of star trackers to 100% when not viewing the sun.

Structural & Field Engineer Intern: Kiewit

May - August 2018, June - August 2019

- Designed supporting structures for large-scale infrastructure projects in Seattle and Los Angeles, utilizing 3D modeling software to optimize designs for cost, efficiency, and constructibility.
- Led the transition to a streamlined CAD review system, improving design accuracy and reducing time-to-build by 1-2 weeks, contributing to faster project approvals and cost control.
- Performed structural calculations and load analyses to assess new and existing formwork, ensuring compliance with engineering standards and safety regulations.
- Designed critical safety and mechanical systems, including fall protection and formwork, enhancing worker safety and structural integrity on active construction sites.
- Assisted in managing a 20+ person construction crew by developing engineering-based work plans and hands-on training programs, ensuring successful implementation of design modifications in the field.

TECHNICAL SKILLS

- **Electronics:** PCB manufacturing & DC circuit design (Eagle CAD), bench top debugging (oscilloscope, multimeter).
- **Programming/Software:** Python, C/C++, bash/CLI. Focus on scientific/data analysis scripting, embedded firmware, and software-defined radio (SDR) radar.
- **Spatial analysis:** GNSS reflectometry, GNSS Field Surveying/Data processing (RTK, PPK, PPP), ArcGIS/QGIS (data processing and visualization), ENVI (remote geospatial data processing).
- **Mathematical/analytical tools:** Matlab, Mathematica, Excel, LabView.
- **3D modeling:** Autodesk, SketchUp and Microstation.
- **Media processing & creation:** Machine Shop (CNC router, 3D printing, lathe, etc.), Advanced proficiency in Adobe Suite.
- **Piloting:** 120+ hours as private pilot, tailwheel and seaplane endorsements.
- **Foreign languages:** Advanced in Spanish, proficient in Portuguese, elementary Mandarin.

ACTIVITIES

- **Outdoor education/exploration:** Wilderness First Responder, Pomona College outdoors club trip leader. Selected summits include El Capitan (via The Nose) and Mt. Rainier (Emmons-Winthrop Route, DC Glacier Route).
- **Vans RV-7 aircraft build:** Assisted the building of an experimental class aircraft with Harvey Mudd College students.